



III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	D	A	A	B	B	A	B	A	C	A	C	C	A	A

16	17	18	19	20	21	22	23	24	25	26	27			
B	B	A	D	B	C	D	D	C	C	B	C			

- 1 B** Magnetic force on YZ

$$F = B \cos \theta \times IL = 0.40 \times \cos 45^\circ \times 5.0 \times 0.12 = 0.17 \text{ N}$$

Using Fleming's LHR, the magnetic force acting on YZ is directed out of the page.
- 2 D** Magnetic force on particle provides for centripetal force for particle to undergo circular motion.

$$Bqv = \frac{mv^2}{r} \Rightarrow B = \frac{mv}{2rq} \propto \frac{1}{rq}$$

So $B' = \frac{r}{r'} \cdot \frac{q}{q'} \cdot B = \frac{2}{1} \cdot \frac{2}{1} \cdot B = 4B$
- 3 A** A charge moving parallel to the magnetic field experiences no magnetic force. So the charge moves in a straight line.
- 4 A** By Fleming's LHR, the force acting on the wire section XY will be towards the left.
- 5 B** Magnetic force on XY must be downwards. Using Fleming's LHR, current flows from Y to X.

Applying principle of moments about the pivot,

$$Zy = BILx$$

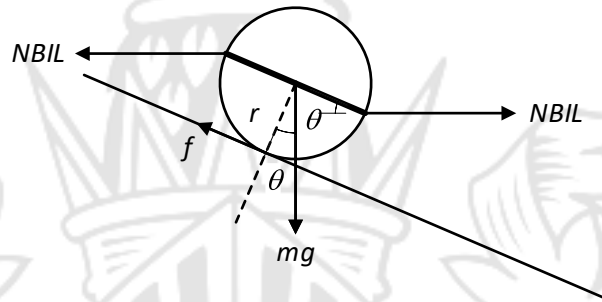
$$\frac{Z}{I} = \frac{BLx}{y} = \frac{0.050 \times 0.50 \times 1.0}{0.50} = 0.050 \text{ N A}^{-1}$$

So if $Z = 0.098 \text{ N}$, then $I = 1.96 \text{ A}$.

- 6 B Between the plates, magnetic force F_B on electron is upward and electric force F_E is downward.
 250 eV electron passes through undeflected as $F_B = F_E$.
 More energetic electrons will impact plate X since $F_B > F_E$. Less energetic electrons will impact plate Y since $F_B < F_E$.

- 7 A Since the electrons are moving out of the page, according to Fleming's LHR, the magnetic force is directed to the right while electric force is directed upwards.

- 8 B



The magnetic forces on the coil is a couple with separation $2r \sin \theta$.

To prevent the cylinder from rolling, the resultant moment about the point of contact is zero.

$$mgr \sin \theta = NBIL \times 2r \sin \theta$$

$$I = \frac{mg}{2NBL} = \frac{0.250 \times 9.81}{2 \times 10 \times 0.50 \times 0.10} = 2.45 \text{ N}$$

Note: In order the cylinder to roll, there must be friction f between the cylinder and the slope.

- 9 A Since the magnetic flux density B is always perpendicular to the coil at any angular displacement θ , the forces (equal in magnitude but opposite in direction) on the each side of the coil ($F = BIL$) is constant. Hence the torque is also a constant.
- 10 C Current along PR is parallel to the magnetic field and so there is no magnetic force on PR.
 According to Fleming's LHR, magnetic force on RQ is directed into the page and magnetic force on QP is directed out of the page.
 These two forces produce a torque that causes the wire frame to rotate about the vertical axis WX.
- 11 A Orbital period $T = 2\pi m/Bq$.
 Orbital radius $r = mv/Bq \propto 1/q$.
 Magnetic force $F_B = Bqv$.
- 12 C Magnetic force on proton is directed upwards, so electric force on proton and hence electric field must be directed downwards.

Using $v = \frac{E}{B}$,

$$E = vB = 2.00 \times 10^7 \times 1.5 = 3.00 \times 10^7 \text{ N C}^{-1}$$

- 13 C** Currents in X and Y are in the same direction. X and Y experience attractive force, i.e., force on Y is towards X. Currents in Y and Z are in opposite directions. Y and Z experience repulsive force, i.e., force on Y towards X. Resultant force is towards X.

- 14 A** Magnetic force

$$F = BIL \sin \theta = 0.040 \times 5.0 \times 0.030 \times \sin 90^\circ = 0.0060 \text{ N}$$

- 15 A** Magnetic flux density B due to the current in the solenoid is along its axis. Since the velocity of the electron is parallel to B , no magnetic force acts on the electron.

- 16 B** By the principle of conservation of energy,
gain in K.E. = work done by electric field

$$\frac{1}{2}mv^2 - 0 = eV$$

$$\Rightarrow v = \sqrt{\frac{2eV}{m}}$$

Since the electron is undeflected in the velocity selector,

$$F_E = F_B$$

$$\Rightarrow \frac{eV}{d} = Bev = Be\sqrt{\frac{2eV}{m}}$$

Rearranging, charge to mass ratio of electron

$$\frac{e}{m} = \frac{V}{2B^2d^2}$$

- 17 B** By right hand grip rule, resultant magnetic flux density at O due to P and R is zero. The magnetic flux density at O due to Q and S is directed towards B.

- 18 A** For wire R to experience zero resultant force

$$\frac{\mu_0 I_P I_R L}{2\pi r_P} = \frac{\mu_0 I_Q I_R L}{2\pi r_Q}$$

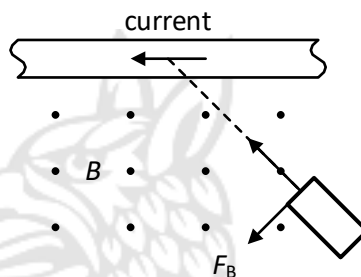
Rearranging,

$$\frac{r_Q}{r_P} = \frac{I_Q}{I_P} = 2$$

That is, wire R must be twice as far from wire Q than from wire P. So, wire R must be placed a distance x to the left of wire P.

Note: The magnitude and direction of the current in wire R is irrelevant.

19 D



In a direction at 90° to the original path away from the wire.

20 B By the principle of moments,

$$F_B \times AS = mg \times SR$$

$$\Rightarrow B \times \frac{2.0}{2.5} \times 0.060 \times 3L = 0.030 \times 9.81 \times 2L$$

$$\Rightarrow B = 4.1 \text{ T}$$

21 C

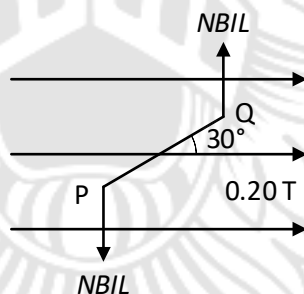


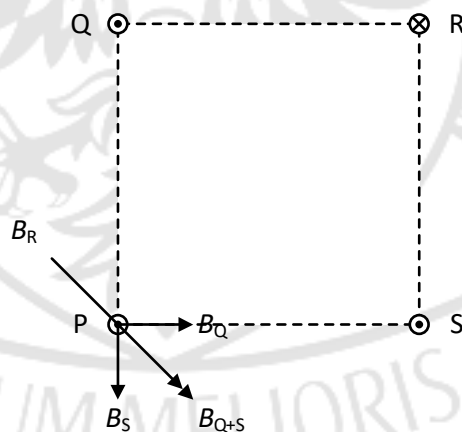
Fig. B

Torque due to the couple

$$\tau = NBIL^2 \cos 30^\circ = 50 \times 0.20 \times 3.0 \times 0.040^2 \cos 30^\circ = 0.042 \text{ N m}$$

22 D Using Fleming's left-hand rule, the magnetic force on the electrons is directed from Y to X. So side X is negatively charged while side Y is positively charged, resulting in an electric field directed from Y to X.

23 D For wire P to experience zero resultant force, the resultant magnetic flux density at wire P must be zero.



The magnetic flux densities due to wire Q, S and R are

$$B_Q = \frac{\mu_0 I}{2\pi d}; B_S = \frac{\mu_0 I}{2\pi d}; B_{Q+S} = \frac{\mu_0 I}{\sqrt{2}\pi d}$$

For zero resultant flux density

$$B_R = B_{Q+S}$$

$$\Rightarrow \frac{\mu_0 I'}{2\pi\sqrt{2}d} = \frac{\mu_0 I}{\sqrt{2}\pi d}$$

$$\Rightarrow I' = 2I$$

- 24 C** Magnetic force on electron provides for centripetal force for the electron to undergo circular motion.

$$Bev = mv\omega$$

$$\Rightarrow Be = m \times \frac{2\pi}{T}$$

$$\Rightarrow T = \frac{2\pi m}{Be} = \frac{2\pi \times 9.11 \times 10^{-31}}{3.0 \times 1.60 \times 10^{-19}} = 1.2 \times 10^{-11} \text{ s}$$

Since the electron only travels half a circle, so the time spent is $6.0 \times 10^{-12} \text{ s}$.

Note: The period T of circular motion is independent of v .

- 25 C** Magnetic force on wire

$$F = BIL \sin \theta = \mu_0 nI \times I_{xy} L_{xy} = 4\pi \times 10^{-7} \times \frac{20000}{1.2} \times 3.0 \times 1.5 \times 0.020 = 1.9 \times 10^{-3} \text{ N}$$

$$= \left(4\pi \times 10^{-7} \times \frac{20000}{1.2} \times 3.0 \right) \times 1.5 \times 2.0 \times 10^{-2}$$

$$= 1.9 \times 10^{-3} \text{ N}$$

- 26 B** For X,

$$B = \frac{\mu_0 \times 100 \times 1.5}{2 \times 0.120} = 625\mu_0 \text{ (directed out of page)}$$

For Y,

$$B = \frac{\mu_0 \times 100 \times 1.0}{2 \times 0.180} = 278\mu_0 \text{ (directed into page)}$$

So, the resultant flux density

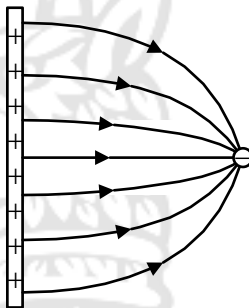
$$B = 625\mu_0 - 278\mu_0 = 347\mu_0 \text{ (directed out of page)}$$

- 27 C** Wire X is attracted towards Y but repelled by Z. However, since XZ is twice as far as XY, the attractive force by Y is stronger than the repulsive force by Z. So the resultant force on X is towards Y.

Structured Questions

1. [ACJC/H2/Prelims 2018/P3/6 Part]

b. i.



Field lines originate from rod and end with point charge.

Field lines are perpendicular to rod.

ii. 1. Magnetic force on charge

$$F = Bqv \sin 90^\circ = 0.010 \times 2.0 \times 10^{-6} \times 0.015 = 3.0 \times 10^{-10} \text{ N}$$

2. According to right-hand grip rule, the magnetic field at the charge is directed into the page.

Using Fleming's LHR, the charge experiences a magnetic force directed to the right.

This magnetic force has varying magnitude (due to non-uniform magnetic field) and is always perpendicular to the velocity of charge. Hence path is curved.

3. Set up a uniform electric field directed to the right (using parallel plates with the right plate at higher potential than the left).

The negative charge will experience an electric force directed to the left. The resultant force on the charge is zero if this electric force is equal in magnitude to the magnetic force.

OR

Set up an identical rod carrying the same current flowing in the same direction such that the charge is at the centre of the two rods.

The resultant magnetic field at the charge is zero. So there is zero magnetic force acting on the charge.

iii. 1. Magnetic flux density

$$B = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \times 0.0040}{2\pi \times 0.20} = 4.0 \times 10^{-9} \text{ T}$$

2. Using right-hand grip rule, the magnetic field due to the coil is directed out-of-page while the magnetic field due to rod at P is directed into the page.

The resultant magnetic flux density at P is the vector sum of two opposing magnetic fields.

3. The segment of coil nearer to the rod experiences a repulsive force (towards right) while the segment further from the rod experiences an attractive force (towards left).

The repulsive force is stronger than the attractive force as the magnetic flux density decreases with distance from the rod, therefore the coil will experience a resultant force directed to the right.

2. [AJC/H2/Prelims 2018/P3/7 Part]

- c. i. The increased reading on the top-pan balance implies an additional downward force on the magnet due to current-carrying wire.

According to N3L, the magnet exerts an upward force on the wire.

By Fleming's LHR, we can deduce that pole P is a north pole.

- ii. The increased apparent weight is the magnetic force acting on the magnet.

$$BIL = mg \Rightarrow B = \frac{mg}{IL} = \frac{0.0023 \times 9.81}{2.6 \times 0.044} = 0.20 \text{ T}$$

- iii. The peak current is $I_0 = I_{\text{rms}} \sqrt{2} = 3.7 \text{ A}$. This will give rise to a fluctuation of 3.3 g.

The reading on the scale will range from $m \pm 3.3 \text{ g}$ where m is the mass of the magnet.

3. [CJC/H2/Prelims 2018/P2/6]

- a. It is the force per unit length acting on a long straight wire carrying a unit electric current at right angles to the uniform magnetic field.

- b. i. Magnetic flux density at P due to WX

$$B_{\text{WX}} = \frac{\mu_0 I_{\text{WX}}}{2\pi r} = \frac{4\pi \times 10^{-7} \times 9.0}{2\pi \times 50 \times 10^{-3}} = 3.6 \times 10^{-5} \text{ T}$$

Note: B_{WX} is directed along CB.

- ii. Magnetic flux density at P due to YZ

$$B_{\text{YZ}} = \frac{\mu_0 I_{\text{YZ}}}{2\pi r} = \frac{4\pi \times 10^{-7} \times 12.0}{2\pi \times 50 \times 10^{-3}} = 4.8 \times 10^{-5} \text{ T}$$

Resultant magnetic flux density at P

$$B_R = \sqrt{(4.8 \times 10^{-5})^2 + (3.6 \times 10^{-5})^2} = 6.0 \times 10^{-5} \text{ T}$$

Note: B_{YZ} is directed along XW.

4. [CJC/H2/Prelims 2018/P3/5]

- a. i. Magnetic force $F = Bqv$ is directed downward.
- ii. The magnetic force acting on the electron is always perpendicular to its velocity. So The acceleration is only in the radial direction which changes the direction of its velocity but not its magnitude (speed).

Furthermore, the magnitude of the magnetic force is constant as speed of the electron is constant. This means that the radius of circular motion is constant.

- b. i. Gain in K.E. is work done by electric field

$$\frac{1}{2}mv^2 = qV \Rightarrow v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 5000}{9.11 \times 10^{-31}}} = 4.19 \times 10^7 \text{ m s}^{-1}$$

- iii. 1. Magnetic force provides for the centripetal force

$$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{Bq} = \frac{9.11 \times 10^{-31} \times 4.19 \times 10^7}{2.0 \times 10^{-3} \times 1.6 \times 10^{-19}} = 0.12 \text{ m}$$

2. The energetic electrons bombard the atoms along the circular path to cause excitation. These excited atoms de-excite by emitting photons and hence the circular path is visible.

5. [DHS/H2/Prelims 2018/P2/6]

- a. The magnetic field due to current in wire B is directed into the paper. By Fleming's LHR, this resulted in a magnetic force on wire A directed towards wire B.

By N3L, wire A exerts an equal but opposite force (towards wire A) on wire B. These forces are mutually attractive in nature.

- b. i. Magnetic flux density at P

$$B = \frac{\mu_0 I}{2\pi d} = \frac{4\pi \times 10^{-7} \times 90.0}{2\pi \times 0.500} = 3.60 \times 10^{-5} \text{ T}$$

Magnetic force provides for centripetal force.

$$Bqv = \frac{mv^2}{r}$$

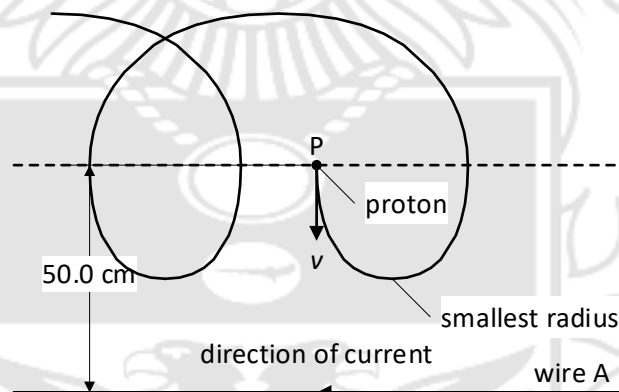
$$r = \frac{mv}{Bq} = \frac{1.67 \times 10^{-27} \times 1.00 \times 10^3}{3.60 \times 10^{-5} \times 1.60 \times 10^{-19}} = 0.290 \text{ m}$$

Note: The path of the proton is not circular as the magnetic field in the region is not uniform.

- ii. As proton moves nearer to wire A, magnetic flux density increases as $B \propto 1/r$ and radius of circular motion decreases ($r = \frac{mv}{Bq}$).

Eventually at the nearest location to the wire, the velocity of proton is parallel to wire, therefore magnetic force is the greatest and is directed away from wire. At this position, the radius is the smallest.

Thereafter, the proton moves in a path that brings it further away from the wire. As the proton moves away from the wire, the magnetic flux density B decreases and radius increases.



Note: Only the path between P and wire A is required. The path above and beyond P is shown for your information.

6. [EJC/H2/Prelims 2018/P2/3]

- a. One tesla is the uniform magnetic flux density which acting normally to a long straight wire carrying a current of 1 ampere, causes a force per unit length of 1 N m^{-1} on the wire.
- b. i. Charged particles moving in a magnetic field experience a magnetic force that is perpendicular to its velocity. By Newton's 2nd law, there is no acceleration parallel to its velocity (acceleration is radial), its speed remains constant at 4500 m s^{-1} .
- ii. Magnetic force provides for centripetal force.

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq} = \frac{2.66 \times 10^{-26} \times 4500}{2.00 \times 10^{-3} \times 1.60 \times 10^{-19}} = 0.374 \text{ m}$$

- iii. Angular velocity

$$\omega = \frac{v}{r} = \frac{4500}{0.374} = 12000 \text{ rad s}^{-1}$$

- iv. Angular displacement

$$\theta = \sin^{-1}\left(\frac{OY}{r}\right) = \sin^{-1}\left(\frac{0.200}{0.374}\right) = 0.564 \text{ rad}$$

- v. 1. To remain in the field without exiting XY,

$$r = \frac{mv}{Bq} = \frac{p}{Bq} \leq 0.200 \text{ m}$$

$$p \leq 0.200 \times 2.00 \times 10^{-3} \times 1.60 \times 10^{-19} = 6.40 \times 10^{-23} \text{ kg m s}^{-1}$$

2. Magnetic force provides for centripetal force.

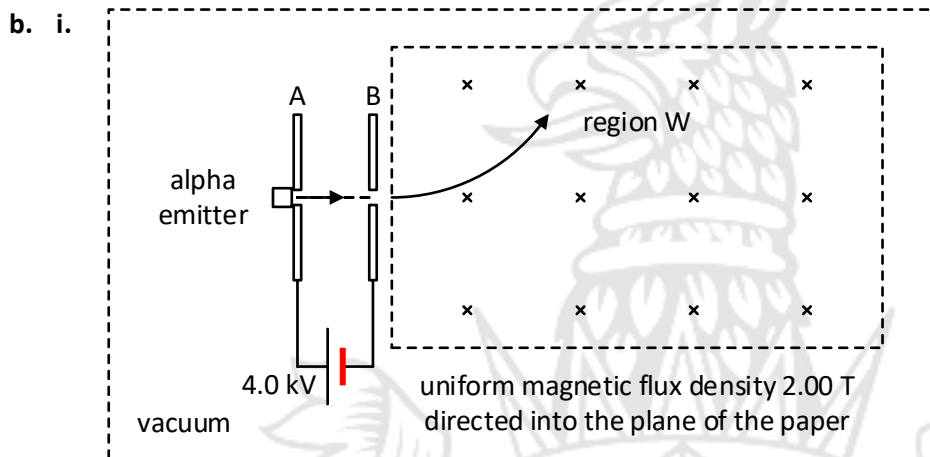
$$Bqv = mv\omega \Rightarrow \frac{Bq}{m} = \omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi m}{Bq}$$

To complete a semicircle

$$t = \frac{T}{2} = \frac{\pi m}{Bq}$$

7. [IJC/H2/Prelims 2018/P3/8]

- a. One tesla is the uniform magnetic flux density which acting normally to a long straight wire carrying a current of 1 ampere, causes a force per unit length of 1 N m^{-1} on the wire.



- ii. Alpha particle moving in a magnetic field experience a magnetic force that is perpendicular to its velocity. No work is done by the magnetic force and hence kinetic energy of the alpha particles remains constant.
- iii. Gain in K.E. is work done by electric field

$$\frac{1}{2}mv^2 = qV \Rightarrow v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2 \times 3.20 \times 10^{-19} \times 4000}{6.68 \times 10^{-27}}} = 6.2 \times 10^5 \text{ m s}^{-1}$$

Note:

Mass of alpha particle is approximately $4m_p = 4 \times 1.67 \times 10^{-27} = 6.68 \times 10^{-27} \text{ kg}$.

Charge of alpha particle is $3.20 \times 10^{-19} \text{ C}$.

- iv. For alpha particle to be undeflected

$$F_B = F_E \Rightarrow Bqv = qE$$

$$E = vB = 6.2 \times 10^5 \times 2.00 = 1.2 \times 10^6 \text{ V m}^{-1}$$

Since magnetic force is directed upwards, electric force must be directed downwards. So electric field strength is directed downwards.

- c. i. Magnetic force provides for centripetal force.

$$Bqv_{\perp} = \frac{mv_{\perp}^2}{r} \Rightarrow r = \frac{mv_{\perp}}{Bq} = \frac{6.68 \times 10^{-27} \times 6.2 \times 10^5 \sin 30^\circ}{2.00 \times 3.20 \times 10^{-19}} = 3.2 \times 10^{-3} \text{ m}$$

- ii. Magnetic force provides for centripetal force.

$$Bq_{\alpha}v_{\perp} = m_{\alpha}v_{\perp}\omega \Rightarrow \frac{Bq_{\alpha}}{m_{\alpha}} = \omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi m_{\alpha}}{Bq_{\alpha}}$$

- iii. Pitch $p = v_{\parallel}T = 6.2 \times 10^5 \cos 30^\circ \times \frac{2\pi \times 6.68 \times 10^{-27}}{2.00 \times 3.20 \times 10^{-19}} = 0.0352 \text{ m}$

- iv. Specific charge of positron $\gg$ specific charge of alpha particle.

$$\frac{q_{e^+}}{m_{e^+}} = \frac{1.60 \times 10^{-19} \text{ C}}{9.11 \times 10^{-31} \text{ kg}} = 1.76 \times 10^{11} \text{ C kg}^{-1}$$

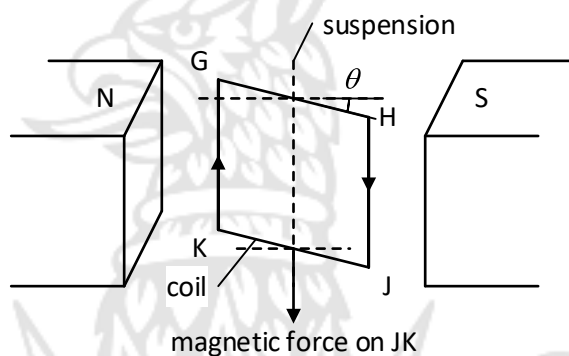
$$\frac{q_{\alpha}}{m_{\alpha}} = \frac{3.20 \times 10^{-19} \text{ C}}{6.68 \times 10^{-27} \text{ kg}} = 4.79 \times 10^7 \text{ C kg}^{-1}$$

Since period T and radius r are inversely proportional to the specific charge of the particle (with the same initial velocity), the period is shorter and the radius of the helical path is smaller for the positron.

Since pitch p is proportional to period T , the pitch of the helical path will also be shorter.

8. [JJC/H2/Prelims 2018/P3/9 Part]

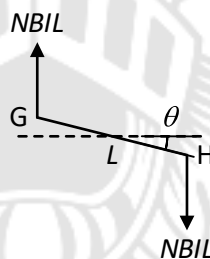
- a. It is the force per unit length acting on a long straight wire carrying a unit electric current at right angles to the uniform magnetic field.
- b. i.



- ii. Magnetic force on JK

$$F = NBIL \sin \theta = 500 \times 2.0 \times 10^{-4} \times 0.40 \times 0.050 \sin 35^\circ = 1.1 \times 10^{-3} \text{ N}$$

- iii. Magnetic force acting on sides HJ and GK is $NBIL$.

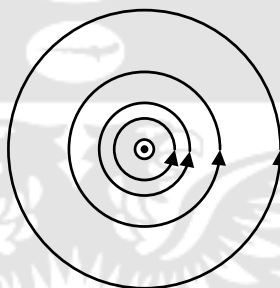


Torque of the couple

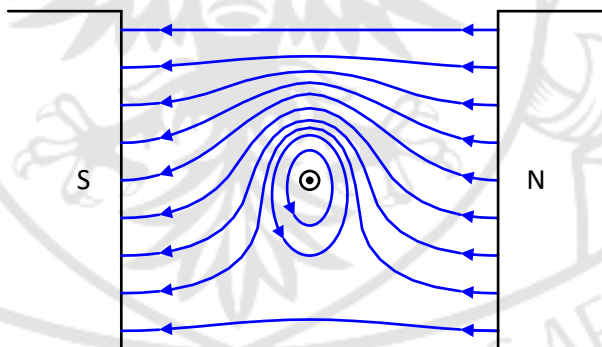
$$\tau = NBIL \times L \cos \theta = 500 \times 2.0 \times 10^{-4} \times 0.40 \times 0.050^2 \cos 35^\circ = 8.2 \times 10^{-5} \text{ N m}$$

9. [MI/H2/Prelims 2018/P3/5 Modified]

- a. i.



- ii.



- b. By Fleming's left-hand rule, there is a downward force on the wire by the magnet. By Newton's third law, there is an equal upward force on the magnet by the wire. Therefore, the balance reading decreases.

10. [MJC/H2/Prelims 2018/P3/8 Part Modified]**b. i. Acceleration of electron**

$$a_y = \frac{F_E}{m_e} = \frac{eE}{m_e} = \frac{1.60 \times 10^{-19} \times 4000}{9.11 \times 10^{-31}} = 7.025 \times 10^{14} = 7.03 \times 10^{14} \text{ m s}^{-2}$$

ii. Consider the horizontal motion. Time taken to transverse the E-field

$$t = \frac{s_x}{u_x} = \frac{0.051}{1.70 \times 10^7} = 3.00 \times 10^{-9} \text{ s}$$

Consider the vertical motion. Vertical component of velocity

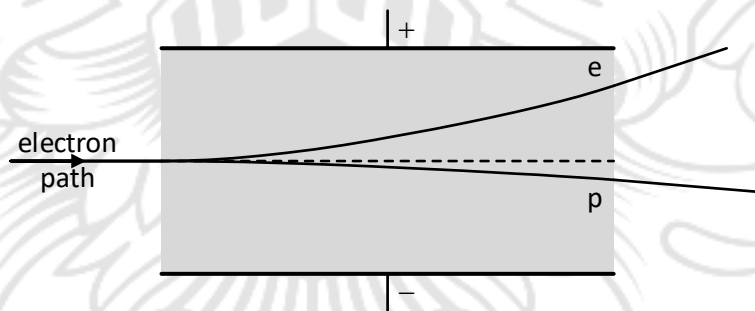
$$v_y = u_y + a_y t = 0 + 7.03 \times 10^{14} \times 3.00 \times 10^{-9} = 2.11 \times 10^6 \text{ m s}^{-1}$$

Final velocity of electron

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(1.70 \times 10^7)^2 + (2.11 \times 10^6)^2} = 1.71 \times 10^7 \text{ m s}^{-1}$$

Amount of deflection

$$\theta = \tan^{-1} \frac{v_y}{v_x} = \tan^{-1} \frac{2.11 \times 10^6}{1.70 \times 10^7} = 7.08^\circ$$

iii.

Proton is deflected in the opposite direction as it has opposite charge to electron.

Proton experiences the same magnitude of force as it has the same magnitude of charge as an electron. However, with a much greater mass, proton has a much lower acceleration so is deflected to a lesser degree.

c. The magnetic force on the electron is constant in magnitude and always perpendicular to its velocity. So, the electron constantly changes its direction of motion but not its speed. Hence the electron travels in a circular path.

The electric force on the electron is constant in magnitude and direction. So, the electron is accelerated only vertically and not horizontally. Hence the electron travels in a parabolic path.

11. [NJC/H2/Prelims 2018/P3/4]

a. i.



ii. With a lower speed, the alpha particles experience a lower magnetic force F_B while the electric force F_E remains unchanged. So, the alpha particles are deflected to the right.

b. Magnetic force on alpha particle provides for the centripetal force for the alpha particle to undergo circular motion.

$$Bqv = \frac{mv^2}{r}$$

Distance between point of entry and impact

$$d = 2r = \frac{2mv}{Bq} = \frac{2 \times 4 \times 1.67 \times 10^{-27} \times 1.00 \times 10^5}{0.050 \times 3.20 \times 10^{-19}} = 8.35 \times 10^{-2} \text{ m}$$

c. Stage 1: Velocity selector

Equation: $v = \frac{E}{B}$

Since the speed of beta particles is 100 that of alpha particles, for beta particles to remain undeflected would require an electric field 100x stronger or magnetic field 100x weaker.

Stage 2: Ion separator

Equation: $r = \frac{mv}{Bq}$

Beta particles are negatively charged and are deflected to the right of entry point. So, the photographic plate must be placed on the right of entry point.

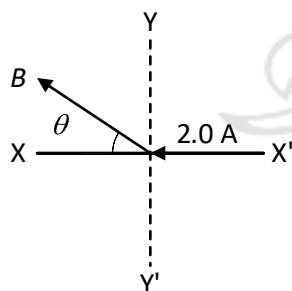
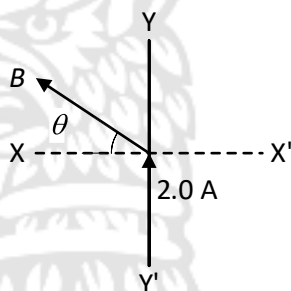
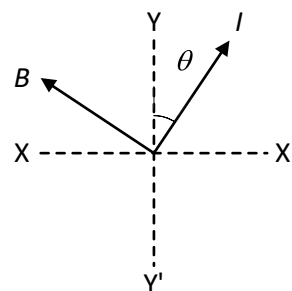
$$v_e = 100v_\alpha; m_e = \frac{1}{7200}m_\alpha$$

Since $B \propto \frac{mv}{q}$, for beta particles to have the same distance between point of entry and impact, the magnetic flux density of

$$B' = \frac{m_e}{m_\alpha} \cdot \frac{v_e}{v_\alpha} \cdot \frac{q_\alpha}{q_e} \cdot B = \frac{1}{7200} \cdot 100 \cdot 2 \cdot 0.050 = 1.4 \times 10^{-3} \text{ T}$$

12. [NYJC/H2/Prelims 2018/P2/7 Modified]

- a. Magnetic flux density is defined as the force acting per unit length per unit current on a current-carrying conductor placed at right angles to the magnetic field.
- b. i.

**Fig. 12.1a****Fig. 12.1b****Fig. 12.1c**

In Fig. 12.1a,

$$F_1 = BIL \sin \theta = 0.50 \text{ N} \quad \text{①}$$

In Fig. 12.1b,

$$F_2 = BIL \cos \theta = 0.80 \text{ N} \quad \text{②}$$

① ÷ ②:

$$\tan \theta = \frac{5}{8}$$

$$\Rightarrow \theta = 32^\circ$$

Magnetic flux density

$$B = \frac{F_1}{IL \sin \theta} = \frac{0.50}{2.0 \times 1.0 \times \sin 32^\circ} = 0.47 \text{ T}$$

- ii. As shown in Fig. 12.1c, the wire experiences a maximum magnetic force when it is perpendicular to the B-field. So the wire should be align along 32° east of north.
- c. Distance from wire with the same magnitude of B-field
- $$r = \frac{\mu_0 I}{2\pi B} = \frac{4\pi \times 10^{-7} \times 2.0}{2\pi \times 0.47} = 8.5 \times 10^{-7} \text{ m}$$

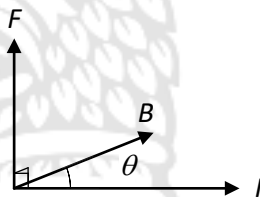
This point is much smaller than the radius of the wire, so it is not possible to find a point with the same magnitude of the B-field.

13. [PJC/H2/Prelims 2018/P2/6]

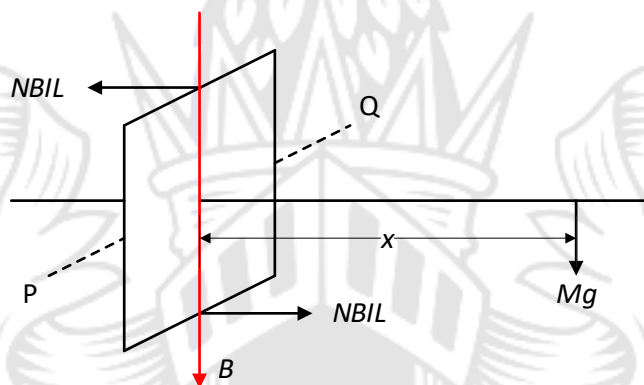
- a. i. Magnetic flux density

$$B = \frac{F}{IL \sin \theta}$$

ii.



- b. i.



The weight of the rider exerts a clockwise torque about the axis PQ. Hence, we can deduce that the forces acting on the square coil must exert an anti-clockwise torque about the same axis.

$$NBIL \times L = Mg x$$

$$\Rightarrow B = \frac{Mgx}{IL^2 N}$$

- ii. Rearranging,

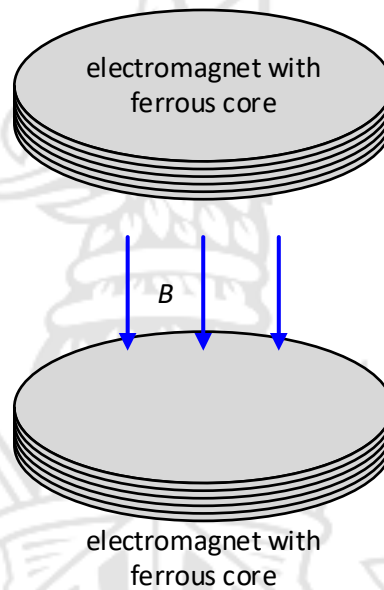
$$x = \frac{NBIL^2}{Mg} \propto IL^2$$

With sides of $L/2$, the length and resistance of wire in the coil is halved. The current I flowing in the coil is doubled since e.m.f. of cell is constant. So $x \propto IL^2$ is halved.

14. [RVHS/H2/Prelims 2018/P3/4 Modified]

- a. Magnetic field is a region of space where a magnetic pole (or magnet), moving charged particle or current-carrying conductor will experience a magnetic force when placed in the field.

b.



- c. i. Magnetic force provides for the centripetal force for the alpha particles to move in a circular path.

$$Bqv = \frac{mv^2}{r}$$

$$\Rightarrow r = \frac{mv}{Bq} = \frac{4u \times v}{B \times 2e} = \frac{2uv}{Be}$$

Note: Mass of alpha particle is approximately $4u$.

- ii. Alpha particles are accelerated by the electric field in the gap (between dees) due to potential difference. So, alpha particles have higher speed and a larger radius ($r \propto v$) of circular path after crossing the gap.

- d. Magnetic flux density at centre of solenoid

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times \left(\frac{3100}{0.39} \right) \times 200 = 1.998 = 2.0 \text{ T}$$

- e. Magnetic force provides for centripetal force for the alpha particles to move in a circular path.

$$Bqv = mv\omega = mv \times \frac{2\pi}{T}$$

Period of circular motion

$$T = \frac{2\pi m}{Bq} = \frac{2\pi \times 4 \times 1.66 \times 10^{-27}}{2.0 \times 2 \times 1.60 \times 10^{-19}} = 6.52 \times 10^{-8} \text{ s}$$

Alpha particles are accelerated twice in each period. So, the total number of times each alpha particle is accelerated

$$N = 2 \times \frac{0.52 \times 10^{-3}}{6.52 \times 10^{-8}} = 15950$$

For alpha particle crossing the gap each time, by the principle of conservation of energy,
work done by electric field = gain in kinetic energy

$$Nq\Delta V = \Delta E$$

$$\Rightarrow \Delta V = \frac{\Delta E}{Nq} = \frac{16 \times 10^6 \times 1.60 \times 10^{-19}}{15950 \times 2 \times 1.60 \times 10^{-19}} = 502 \text{ V}$$

- a. Since the particle is travelling in a clockwise circular path, according to Fleming's left-hand rule, the particle must have a positive charge. , it is a proton.
- b. i. If the frequency is not set accurately, charged particles may experience deceleration when crossing the gap and it will not be possible to attain high speed as desired.
- ii. Magnetic force provides for the centripetal force for the charged particle to move in a circular path.

$$Bqv = mv\omega = mv \times 2\pi f$$

Rearranging,

$$f = \frac{Bq}{2\pi m} = \frac{1.7 \times 1.60 \times 10^{-19}}{2\pi \times 1.67 \times 10^{-27}} = 2.59 \times 10^7 = 2.6 \times 10^7 \text{ Hz}$$

- c. i. Magnetic force provides for the centripetal force for the charged particle to move in a circular path.

$$Bqv = \frac{mv^2}{r}$$

$$\Rightarrow v = \frac{Bqr}{m}$$

Final kinetic energy of particle

$$E_k = \frac{1}{2}mv^2 = \frac{B^2 q^2 r^2}{2m} = \frac{1.7^2 \times (1.60 \times 10^{-19})^2 \times 0.25^2}{2 \times 1.67 \times 10^{-27}} = 1.4 \times 10^{-12} \text{ J}$$

- ii. In 10000 revolutions, the proton crosses the gap and is accelerated 20000 times.

For proton crossing the gap each time, by the principle of conservation of energy,

work done by electric field = gain in kinetic energy

$$q\Delta V = \Delta E$$

$$\Rightarrow \Delta V = \frac{\Delta E}{q} = \frac{1.4 \times 10^{-12}}{20000 \times 1.60 \times 10^{-19}} = 440 \text{ V}$$

16. [VJC/H2/Prelims 2018/P3/4 Modified]

- a. The magnetic force must be directed vertically downward to balance the moment due to weight of mass. Using Fleming's left-hand rule, we can deduce that the current flows from B to A.

- b. Taking moments of force about the pivot,

$$BIL \times 0.080 = mg \times 0.070$$

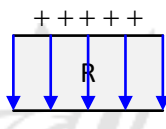
$$\Rightarrow B = \frac{mg \times 0.070}{IL \times 0.080} = \frac{2.0 \times 10^{-3} \times 9.81 \times 0.070}{3.0 \times 6.0 \times 10^{-2} \times 0.080} = 9.5 \times 10^{-2} \text{ T}$$

- c. A non-horizontal magnetic field B can be resolved into its horizontal and vertical components.

The vertical component of B produces a horizontal force on AB and a zero moment of force about the pivot. So the magnitude of the vertical component of B and hence B cannot be determined.

17. [YJC/H2/Prelims 2018/P3/6 Part Modified]

a.



b. To produce a narrow beam of ions.

c. i. Magnet field is directed into the page in region R and out of the page in region S.

ii. When ions pass through the velocity selector, the electric force (downward) on the ions is equal in magnitude but opposite in direction to the magnetic force (upward), i.e., $qE = Bqv$.

This means that ions with speed $v = E/B$ will pass through undeflected and this is independent of mass.

iii. Slower moving ions will experience the same electric force but a lower magnetic force. A downward resultant force will deflect these ions towards the lower (negative) plate.